

Influence of roots on soil preferential paths for infiltration in typical vegetation of the Loess Plateau

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ABSTRACT

Previous studies have indicated that preferential flow is likely to occur during infiltration on the Loess Plateau, with root systems being a key factor. Given the large-scale vegetation construction on the Loess Plateau, it is essential to investigate how root system changes influence preferential path formation. This study investigated the effects of roots of different diameter classes on preferential paths using in situ dye tracer experiments and profile imaging methods. Results showed that the cover of forest vegetation increased the density (root length density, root surface density and root volume density) of roots in soil. The total number of preferential paths in the studied 0–30 cm soil layer was 2524 in the natural secondary forest, which was 1.39 times higher than in the *Pinus tabulaeformis* plantation and 1.48 times higher than in the mixed *Robinia pseudoacacia*-*P. tabulaeformis* plantation. Preferential paths were predominantly distributed in the 0–15 cm soil layer. In forest soil, the maximum number of preferential paths consistently occurred at 5 cm depth, with measured values of 803 in *P. tabulaeformis* plantation, 568 in mixed *R. pseudoacacia*-*P. tabulaeformis* plantation, and 996 in natural secondary forest. By contrast, grassland soil exhibited the greatest number of preferential paths at 10 cm depth (1181 paths) and a lower number of preferential paths at 5 cm depth (414 paths). Most preferential paths exhibited an infiltration aperture radius below 5 mm. Roots of different diameter classes promoted the formation of preferential paths, and the effect of fine roots was more evident than that of coarse roots. Vegetation restoration on the Loess Plateau has altered infiltration patterns, necessitating attention to preferential flow, and root configuration and density critically influence preferential paths development, with implications for forest and hydrological management in this region.

1. Introduction

The Loess Plateau has a fragile ecological environment with severe soil erosion, prompting the implementation of numerous long-term vegetation restoration projects (Jia and Zhu, 2020; Wang et al., 2023). Vegetation restoration has a crucial influence on eco-hydrological processes in the Loess Plateau (Liu et al., 2025). An increase in vegetation cover significantly affects rainfall redistribution processes, such as increasing canopy interception, reducing runoff flushing, and influencing water infiltration (Chen et al., 2024). Vegetation restoration also alters soil permeability and water retention capacity by altering soil conditions (e.g. elevated soil organic matter content and improved aggregate stability) (Menz et al., 2013; Qiu et al.,

2021). Therefore, understanding the effects of vegetation restoration on eco-hydrology helps in elucidating the regional hydrological cycle and water balance in the Loess Plateau under global environmental change.

Amongst the various ecohydrological aspects, infiltration, which refers to the process by which water enters the soil from the soil surface, is an essential process of the hydrological cycle and an influential factor in soil moisture storage and vegetation growth (Cui et al., 2019). Vegetation restoration has significantly increased soil heterogeneity, causing a proportion of precipitation to infiltrate not through the traditionally perceived homogeneous pattern but rather as preferential flow (Li et al., 2007) (Fig. 1). When preferential flow, a rapid water movement infiltrates, it bypasses the soil matrix area and penetrates deep into the soil, thereby affecting vegetation water uptake to a certain

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extent (Weiler and Flühler, 2003; Weiler and McDonnell, 2007).

Preferential flow is prevalent in woodland soil and has become a key factor affecting vegetation recovery and forest management (Valtera et al., 2023). This phenomenon results from the numerous preferential paths in the soil, which enable rapid water transport (Zhang and Cao, 2021). Only a few studies have examined preferential paths, despite their prevalence in soil (Guan et al., 2023b). Crustal movement creates vertical joints in the soil, forming conduits for rapid water transport (Zhao et al., 2020). Meanwhile, evaporation and water loss processes similarly lead to crack formation (Ma et al., 2015). In addition, the interspersed distribution of gravels in the soil creates a fissure network structure (Yi et al., 2021), and the plant root system forms a complex 3D structure during its growth (Maier et al., 2019). Accordingly, numerous macropores form in the soil. These fissures and macropores generate numerous preferential paths of varying aperture sizes in the soil, causing the occurrence of preferential flow.

The research team has previously conducted extensive investigations into the preferential flow in the Loess Plateau woodlands and selected numerous environmental factor indicators, identifying vegetative roots in the soil the key factor influencing the degree of preferential flow development (Guan et al., 2024a). The Loess Plateau has deep soil with a low gravel content (Finke et al., 2013), making plant root systems a key factor in the formation of preferential paths. After a long period of vegetation restoration, interactions between the vegetation root systems and the soil greatly alter soil pore properties and influence infiltration (Mair et al., 2022). During root growth and decay, gaps between roots and soil and large pores formed by root pressure on the soil can create preferential paths (Wu et al., 2020). In addition, carbon and nitrogen sources, from those provided by root apoplasts and secretions, promote the activities of soil fauna and microorganisms, which influence the soil pore structure and facilitate the formation of preferential paths (Jačka et al., 2021).

Research on the effects of roots in the Loess Plateau soil on preferential flow development and variability has achieved some progress (Lu et al., 2024; Niu et al., 2024). However, studies that specifically focus on preferential paths in this region remain unexplored, hindering an extensive understanding and mechanistic explanation of preferential flow. Although vegetative root systems contribute to the formation of preferential paths, the extent to which different root geometries influence preferential paths varies across environments (Wu et al., 2021). Previous research suggested that the effects of thick and thin roots on preferential paths are not identical (Nespoulous et al., 2021). How different vegetation restoration types affect preferential paths remain unclear, particularly how root morphology influences the number and aperture size of preferential paths. This study will determine the

number, aperture size and distribution of preferential paths on stained soil profiles by tracing preferential flow with dye solutions, combined with an analysis of root geometry at different diameters. This combined approach allows the study to fundamentally investigate how vegetation restoration affects water infiltration and preferential flow formation.

The influence of root systems on preferential paths in the context of vegetation restoration on the Loess Plateau cannot be ignored. To gain a deeper understanding of the specific contribution of the roots to the preferential paths, this study described the root characteristics of typical restoration types in the Loess Plateau by analysing the root geometry at different diameter levels. The effects of root systems on preferential paths were resolved through field staining tracer experiments. Our objectives were as follows: (i) to quantify and analyse the distribution and characteristics of root systems in typical vegetation restoration types; (ii) to quantify and analyse the number, aperture size and distribution of preferential paths in typical vegetation restoration types; and (iii) to clarify the influence and degree of contribution of root systems to the formation of preferential paths.

2. Materials and methods

2.1. Study site

The study area is located in the Caijiachuan watershed of Ji County, Shanxi Province, China (110°39'45"E–110°47'45"E, 36°14'24"N–36°18'23"N), within the semi-humid eastern portion of the Loess Plateau. This study area experiences a continental monsoon climate, with an average annual temperature of 10.2 °C and annual precipitation of 571 mm. The soil is classified as cinnamon soil derived from loess parent material, with a soil layer thickness exceeding 10 m. Two large-scale afforestation campaigns occurred in 1991 and 2001. The study area features extensive natural secondary forests and planted shelterbelts, with primary land use types being forest, cropland, grassland, and orchards. The dominant plantation species include *Robinia pseudoacacia*, *Pinus tabulaeformis*, and *Platycladus orientalis*.

Three typical vegetation types were selected, namely, pure *P. tabulaeformis* plantation (PP), mixed *R. pseudoacacia*–*P. tabulaeformis* plantation forest (MF) and natural secondary forest (SF). The experiments were conducted using the grassland (G) as the control (Fig. 2). Three sample plots were set up in each vegetation type, with a size of 10 m × 10 m. Each sample plot was surveyed to assess the basic vegetation characteristics (Table. 1).

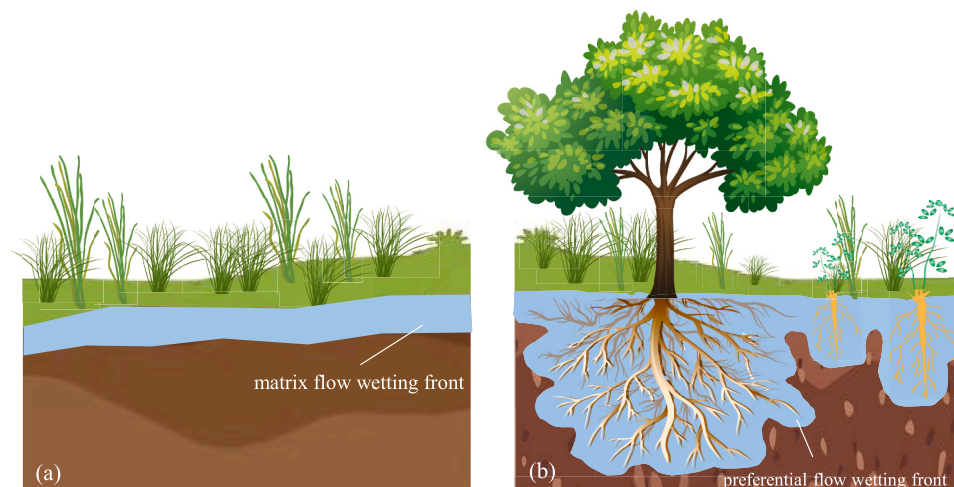


Fig. 1. Effect of vegetation restoration on infiltration. (a) Before and (b) after vegetation restoration.

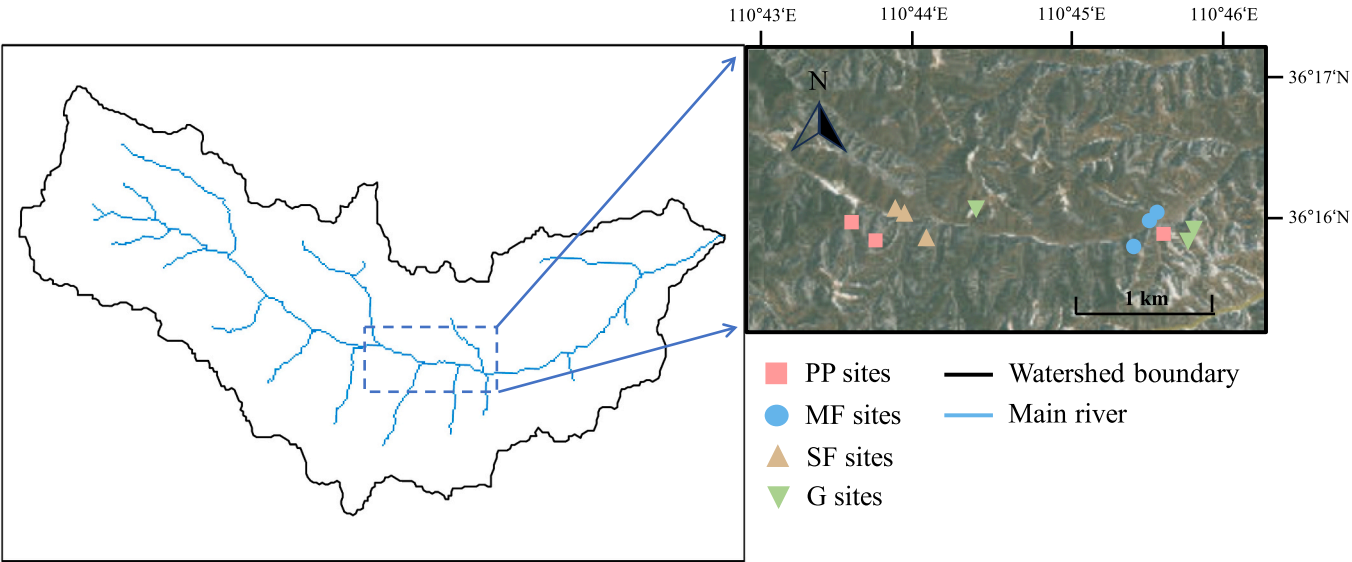


Fig. 2. Study sites.

Table 1
Basic information of the study sites.

Study Site	Elevation (m)	Forest age (years)	Stand density (plants/hm ²)	Crown closure (%)	Main vegetation	Dominant understory vegetation
PP	No.1 1147	34	1850	70	<i>Pinus tabulaeformis</i>	<i>Periploca sepium</i> , <i>Sophora davidia</i> , <i>Rubia cordifolia</i> , etc.
	No.2 1141	34	1550	61		
	No.3 1101	34	1575	65		
MF	No.1 1105	34	1300	65	<i>Pinus tabulaeformis</i> , <i>Robinia pseudoacacia</i>	<i>Rosa xanthines</i> , <i>Artemisia sacrorum</i> , <i>Carex humilis</i> Leyss, etc.
	No.2 1101	34	1650	60		
	No.3 1049	34	1050	52		
SF	No.1 1092	41	750	30	<i>Quercus liaotungensis</i> , <i>Populus lasiocarpa</i>	<i>Rosa xanthines</i> , <i>Artemisia sacrorum</i> , <i>Rubia cordifolia</i> , etc.
	No.2 1057	41	925	45		
	No.3 1050	41	1050	48		
G	No.1 1053	/	/	/	<i>Carex lanceolata</i> Boott	/
	No.2 1003	/	/	/		
	No.3 998	/	/	/		

2.2. Experimental design and data measurement

2.2.1. Infiltration experiment and root sample collection

The infiltration test was carried out using the Brilliant Blue staining tracer method. Three staining tracing sample plots were established for each vegetation type. A spot in each sample plot where the ground was flat and avoided the main roots of the trees was selected as a staining sample point. The staining tracer experiment was conducted based on weather forecasts confirming the absence of precipitation on the experimental day and ± 1 adjacent days. This study used heavy rainfall intensity (> 50 mm/24 h) as the experimental benchmark to facilitate the infiltration of the dye solution into the deeper soil layers, allowing for the observation and extraction of preferential flow paths in relatively deep soil. A 60 mm/24 h rainfall intensity was selected as the experimental standard based on historical meteorological data from the study area, which showed rare occurrences of severe rainstorms (70–100 mm/24 h) and extreme rainstorms (> 100 mm/24 h) in recent years. The calibrated size of the staining observation profile was 60 cm $\times$ 60 cm, with the extent framed with a metal frame. The tracer selected for the experiment was 4 g·L⁻¹ of Brilliant Blue dye. Approximately 22.7 L of Brilliant Blue solution would be required for each staining observation profile, based on the simulated rainfall (60 mm) and 5 % of the actual consumption of the experiment (Guan et al., 2023b).

The staining tracer experiment was carried out by (i) clearing the surface without destroying the soil humus layer and then inserting the

metal frame vertically into the soil at a depth of 5 cm. (ii) The prepared Brilliant Blue staining solution was uniformly sprayed within the framed selected staining observation profiles. (iii) After the spraying was completed, the profiles were covered with a plastic film for 24 h to prevent damage by wildlife. (iv) The staining profiles were excavated after 24 h. Given the unstable moisture transport around the metal frame, the excavation excluded 5 cm of the stained boundary, and the actual dimensions of the stained profiles was 50 cm $\times$ 50 cm. Stained profiles were excavated in layers at 5 cm vertical intervals until no stained patches appeared. During the excavation of each layer of the profile, root samples were simultaneously taken with a root auger for the determination of root properties. Three root samples were taken from each soil layer of the profile. (v) Images were taken after the profiles were excavated. A graduated ruler (ruler) and colour cards were used to correct for image distortion when shooting (Guan et al., 2023b) (Fig. 3).

2.2.2. Soil data collection

Soil samples were taken at 30 cm around each staining tracer experimental site for analysis of soil properties. Ring-knife soil samples were collected at 5 cm intervals within the 0–30 cm soil layer with three replicates sampled per layer. A total of 216 soil samples were collected in this study (nine soil samples were obtained from each layer of each vegetation type).

The soil bulk density and porosity were determined through the cutting ring method (Chen et al., 2021). Prior to sampling, each cutting

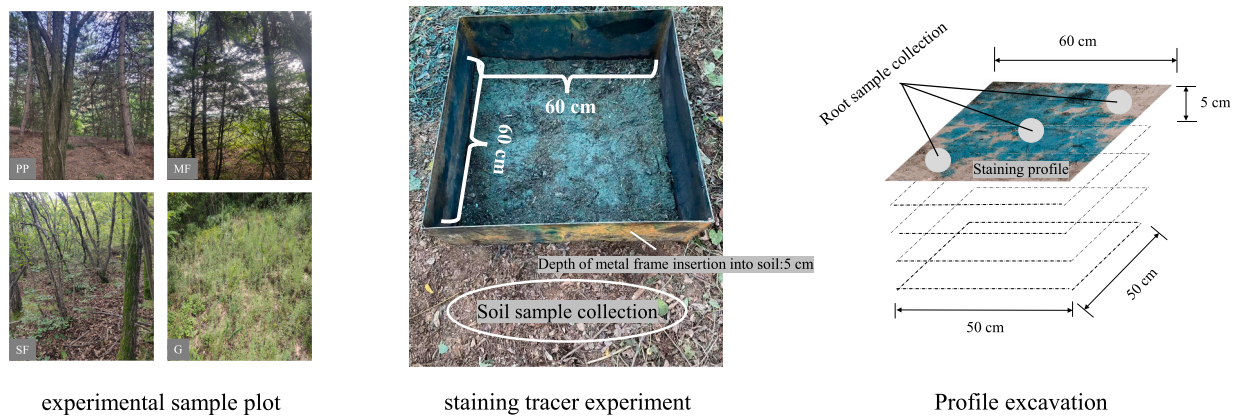


Fig. 3. Schematic of the infiltration experiment.

cylinder's initial weight (W_0 , g) and fixed volume (V , uniformly 106.19 cm^3) were recorded. The cutting cylinder containing the soil sample was weighed in the laboratory (W_{ccs} , g), then placed on a tray filled with distilled water. After 24 h of water absorption, the cylinder with the saturated soil sample was weighed (W_{sat} , g). The cylinder was subsequently transferred to a tray with a dry sand layer at its base, and its weight was recorded after the water in the samples was drained by gravity to constant weight (W_{ws} , g). Finally, the soil sample was oven-dried to constant weight and weighed (W_{dry} , g). Bulk density and porosity were calculated using these measured parameters.

$$\text{Bulk density} = \frac{W_{\text{dry}} - W_0}{V} \quad (1)$$

$$\text{Total porosity} = \frac{W_{\text{sat}} - W_{\text{dry}}}{V} \quad (2)$$

$$\text{Capillary porosity} = \frac{W_{\text{ws}} - W_{\text{dry}}}{V} \quad (3)$$

$$\text{Non-capillary porosity} = \text{Total porosity} - \text{Capillary porosity} \quad (4)$$

The soil organic carbon content was determined by vitriol acid–potassium dichromate oxidation method (Zhao et al., 2022). The soil texture (clay, silt and sand content) was determined by laser diffraction with a Mastersizer 3000, Malvern Instruments (Malvern Panalytical Ltd., UK). The results of the basic soil properties are shown in Table 2.

2.3. Data analysis

2.3.1. Analysis of preferential paths

A stained profile image was processed through Adobe Photoshop 2020 (Adobe Systems Inc., San Jose, CA) to perform geometric correction, light correction, and colour replacement. The stained areas were replaced with black, while the unstained areas were replaced with white to extract pixel points from the image. Noise reduction of colours by separation was performed using Image-Pro Plus 6.0 (Media Cybernetics Inc., Rockville, MD) to obtain accurate black and white images of the stained profiles (Yao et al., 2017).

When water enters the preferential path and infiltrates into the deeper soil layers, a wet perimeter forms around the preferential path. The radius of the stained area can be considered the 'infiltration aperture

Table 2

Basic soil properties of the study sites. The data in the table are the average values of the three plots for each vegetation type.

Vegetation type	Soil depth (cm)	Bulk density ($\text{g}\cdot\text{cm}^{-3}$)	Total porosity (%)	Capillary porosity (%)	Non-capillary porosity (%)	Organic matter content (%)	Soil clay content (%)	Soil silt content (%)	Soil sand content (%)
PP	5	1.00 ± 0.05	50.51 ± 1.40	41.27 ± 2.53	9.25 ± 1.86	1.59 ± 0.54	2.50 ± 0.41	76.72 ± 6.72	20.78 ± 7.05
	10	1.06 ± 0.12	47.46 ± 3.20	38.88 ± 3.11	8.58 ± 3.31	1.31 ± 0.25	2.42 ± 0.34	83.58 ± 2.39	14.00 ± 2.57
	15	1.21 ± 0.09	48.13 ± 1.39	42.85 ± 1.70	5.28 ± 3.30	1.14 ± 0.17	2.66 ± 0.33	84.85 ± 0.96	12.48 ± 1.34
	20	1.19 ± 0.03	48.67 ± 2.13	42.07 ± 1.73	6.60 ± 1.64	0.77 ± 0.13	2.54 ± 0.22	84.10 ± 2.95	13.36 ± 2.82
	25	1.20 ± 0.08	47.78 ± 2.48	41.40 ± 1.77	6.37 ± 1.35	0.79 ± 0.10	2.32 ± 0.41	85.29 ± 0.64	$12.39 \pm 0/97$
	30	1.20 ± 0.07	47.60 ± 1.84	40.68 ± 2.07	6.92 ± 1.47	0.71 ± 0.22	2.58 ± 0.05	82.05 ± 10.65	15.37 ± 10.62
MF	5	0.98 ± 0.10	52.49 ± 5.41	42.51 ± 6.71	9.98 ± 2.31	2.36 ± 0.18	2.67 ± 0.29	78.95 ± 4.78	18.38 ± 5.15
	10	1.14 ± 0.08	49.39 ± 1.79	41.07 ± 2.12	8.33 ± 2.08	1.06 ± 0.39	2.63 ± 0.12	86.04 ± 3.59	11.34 ± 3.71
	15	1.18 ± 0.09	47.50 ± 3.41	42.38 ± 2.79	5.12 ± 2.38	0.71 ± 0.32	2.70 ± 0.57	81.98 ± 12.28	15.32 ± 12.85
	20	1.21 ± 0.05	48.13 ± 1.95	42.73 ± 1.85	5.39 ± 1.06	0.69 ± 0.22	2.67 ± 0.25	80.63 ± 0.91	16.69 ± 0.71
	25	1.14 ± 0.08	47.74 ± 3.40	42.17 ± 3.29	5.57 ± 1.95	0.51 ± 0.12	2.76 ± 0.26	87.31 ± 2.35	9.93 ± 2.24
	30	1.17 ± 0.06	47.85 ± 1.36	43.18 ± 1.54	4.67 ± 1.03	0.48 ± 0.24	2.65 ± 0.58	81.71 ± 1.91	15.63 ± 1.34
SF	5	1.01 ± 0.14	49.32 ± 2.48	39.01 ± 6.05	10.31 ± 4.50	2.08 ± 0.53	1.93 ± 0.25	79.33 ± 3.16	18.72 ± 2.92
	10	1.15 ± 0.1	46.38 ± 1.52	39.36 ± 1.69	7.01 ± 2.41	1.68 ± 0.32	1.9 ± 0.05	78.10 ± 0.34	20.00 ± 0.37
	15	1.15 ± 0.08	47.09 ± 2.08	39.69 ± 2.18	7.40 ± 1.29	1.33 ± 0.22	1.69 ± 0.09	81.04 ± 1.35	17.28 ± 1.23
	20	1.20 ± 0.08	44.60 ± 3.99	38.36 ± 3.45	6.24 ± 1.06	1.10 ± 0.35	1.87 ± 0.21	83.44 ± 0.98	14.69 ± 0.74
	25	1.14 ± 0.07	47.97 ± 4.12	39.51 ± 3.28	8.45 ± 3.14	1.17 ± 0.40	1.8 ± 0.22	80.57 ± 6.83	17.63 ± 7.07
	30	1.15 ± 0.04	46.80 ± 4.23	39.11 ± 5.11	7.69 ± 2.12	1.01 ± 0.41	1.5 ± 0.09	80.79 ± 4.77	17.71 ± 4.75
G	5	1.09 ± 0.12	48.57 ± 2.30	37.85 ± 4.62	10.72 ± 6.22	1.28 ± 0.33	2.17 ± 0.22	81.78 ± 8.11	16.06 ± 8.36
	10	1.16 ± 0.13	43.75 ± 1.37	35.29 ± 2.89	8.46 ± 2.40	0.82 ± 0.26	2.70 ± 0.08	80.94 ± 3.97	16.36 ± 4.05
	15	1.34 ± 0.06	43.15 ± 1.14	38.16 ± 1.52	4.99 ± 1.70	0.71 ± 0.30	2.24 ± 0.21	83.82 ± 0.59	13.94 ± 0.42
	20	1.38 ± 0.08	42.65 ± 1.83	38.85 ± 1.16	3.80 ± 1.05	0.61 ± 0.28	2.31 ± 0.22	86.89 ± 0.21	10.80 ± 0.45
	25	1.25 ± 0.06	44.79 ± 2.27	38.85 ± 2.43	5.93 ± 2.15	0.61 ± 0.28	2.33 ± 0.12	84.76 ± 3.84	12.92 ± 3.80
	30	1.22 ± 0.05	45.15 ± 3.57	38.22 ± 4.32	6.93 ± 3.09	0.67 ± 0.23	2.82 ± 0.74	81.67 ± 8.38	15.52 ± 8.89

radius' (R) of the soil preferential flow path. The preferential paths were extracted and calculated using Image Pro Plus 6.0 (Media Cybernetics Inc., Rockville, MD). Considering the profile of each soil layer as a whole, the staining influence area was divided into several independent patches according to the spatial topological relationship, using the morphological calculation method (Watershed Split). The area of the staining influence area was the number of pixel points occupied by the patches. These patches were approximated as circles, and their radii were calculated based on the staining area. The infiltration aperture radius (R) was classified as $1 \leq R < 2.5$, $2.5 \leq R < 5$, $5 \leq R < 10$ and $10 \leq R$ (R unit: mm) (Zhang et al., 2016) (Fig. 4).

2.3.2. Analysis of roots

The collected roots were carefully rinsed clean. The morphological characteristics of the roots were analysed by WinRHIZO 2020 (Reagent Instruments Inc., Québec, QC, Canada). For calculation, the diameter class D of the roots was divided into four classes, namely, $1 \leq D < 2$, $2 \leq D < 3$, $3 \leq D < 4$ and $4 \leq D$ (D unit: mm). The quantity of the root systems was characterised by root length density, root surface area density and root volume density (Shen et al., 2023).

$$RLD = \frac{RL}{V} \quad (1)$$

where RLD is the root length density ($\text{cm} \cdot \text{cm}^{-3}$), RL is the root length (cm) and V is the soil sample volume (cm^3).

$$RSD = \frac{RS}{V} \quad (2)$$

where RSD is the root surface density ($\text{cm}^2 \cdot \text{cm}^{-3}$) and RS is the root surface area (cm^2).

$$RVD = \frac{RV}{V} \quad (3)$$

where RVD is the root volume density ($\text{mm}^3 \cdot \text{cm}^{-3}$) and RV is the root volume (mm^3).

2.3.3. Analytical method

SPSS 22.0 (SPSS Inc., Version X; IBM, Armonk, NY, USA) was used for statistical analysis and tests for normality and homogeneity of the variance. One-way analysis of variance (ANOVA) and least significant difference (LSD) post hoc tests were used to test for differences in root systems and preferential pathways between vegetation types and soil depths. Pearson correlation analysis was used to express the correlation between the factors. The threshold for statistically significant differences was $P < 0.05$.

3. Results

3.1. Root characteristics of typical vegetation types

The root characteristics varied with different vegetation types and

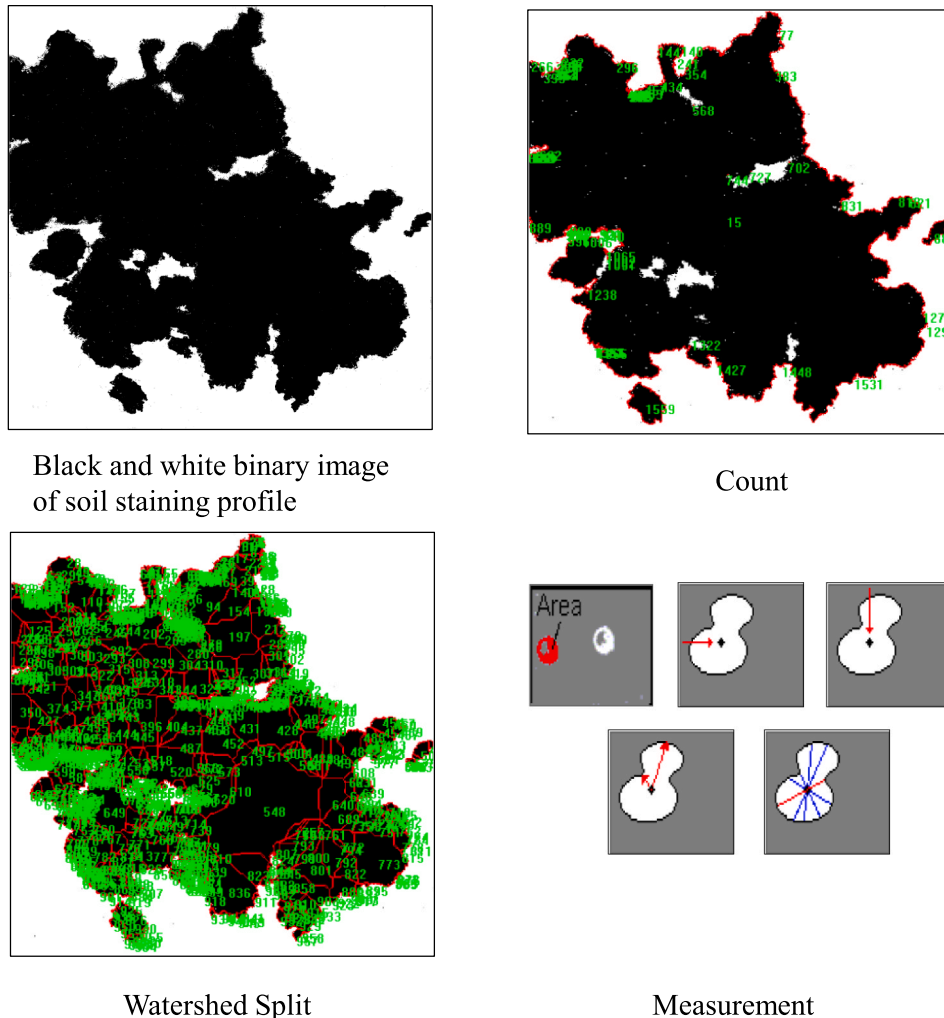


Fig. 4. Schematic of preferential path extraction.

soil depths (Fig. 5). In the four vegetation types (PP, MF, SF and G), the RLD of roots with $D \leq 1$ mm accounted for 83.97 %, 85.20 %, 89.27 %, and 92.81 % of the total RLD, respectively (mean values in 0–30 cm soil), demonstrating significantly higher proportions than the other diameter classes. The RLD of roots with $D > 2$ mm contributed no more than 6 % of the total RLD across all four vegetation types. In RSD, finer roots with smaller diameters ($D \leq 1$ mm) exhibited relatively higher RSD values. The proportion of RSD contributed by these roots with $D \leq 1$ mm

in the 0–30 cm soil layer averaged 57.77 % in PP, 54.28 % in MF, 60.88 % in SF, and 71.31 % in G. In RVD, fine roots did not consistently show higher values than coarse roots. In MF and SF soil, the RVD of roots with $D > 4$ mm accounted for 39.97 % (MF) and 28.81 % (SF) of the total RVD respectively (mean values in 0–30 cm soil), exceeding those of the other diameter classes.

Comparison of the root characteristics of the various vegetation types revealed some regularities and differences between them. Overall,

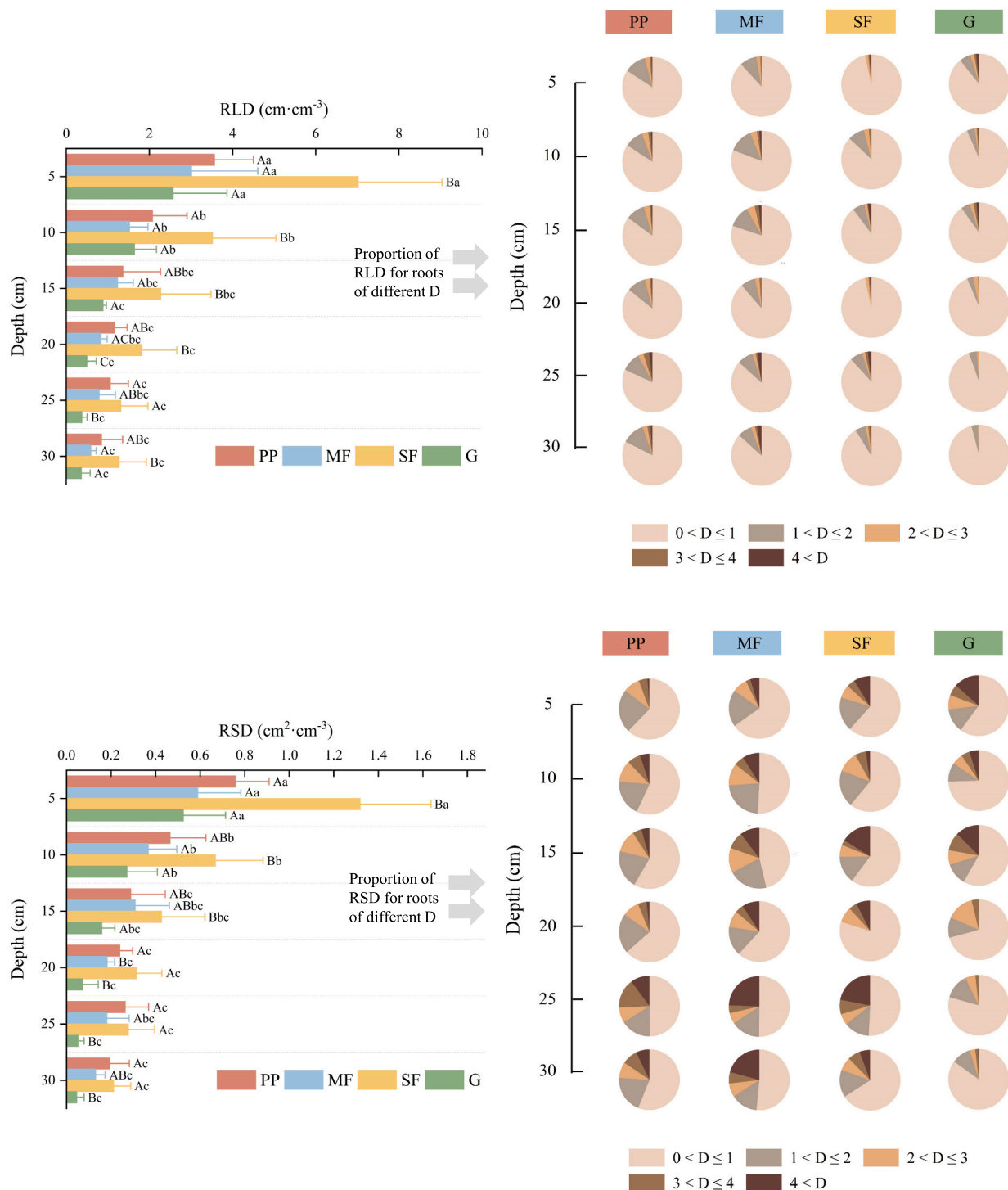


Fig. 5. Root characteristics of typical vegetation types. The significant differences are the results of the variations in RLD (RSD and RVD) under different conditions; the different capital letters represent the significant differences in RLD (RSD and RVD) for the various vegetation types at the same soil depth; the different lowercase letters represent the significant differences in RLD (RSD and RVD) at various soil depths for the same vegetation type.

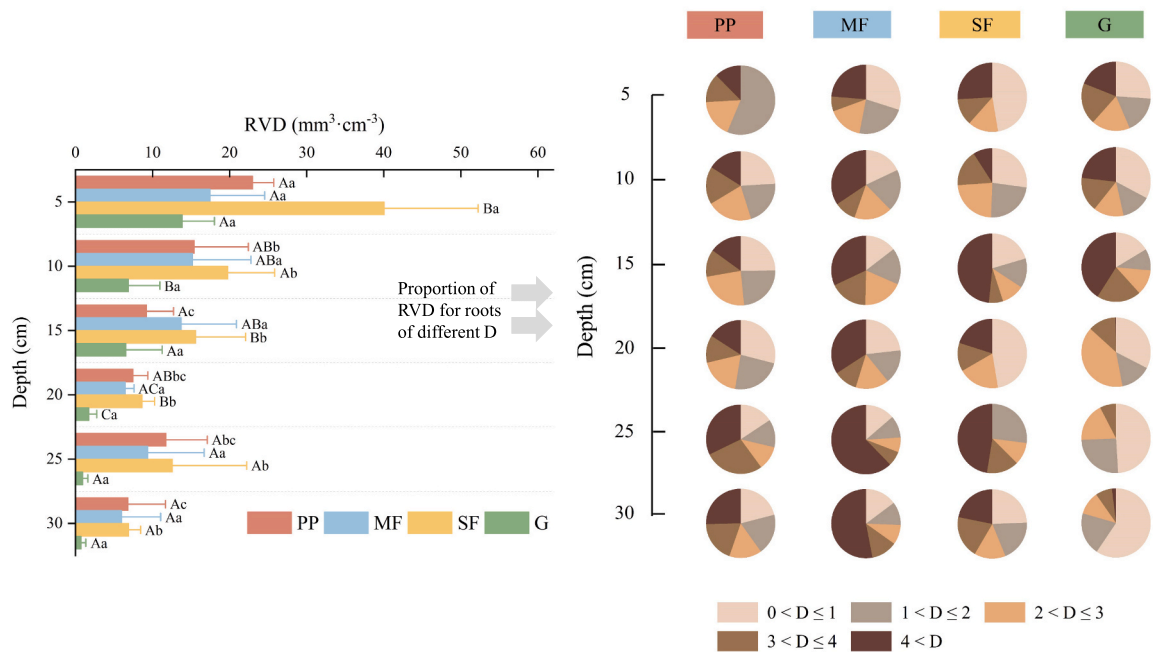


Fig. 5. (continued).

woodland exhibited greater RLD, RSD, and RVD compared with grassland, with a relatively higher coarse root density. Natural secondary forest demonstrated the highest root density parameters amongst the three vegetation restoration types. At all soil depths, natural secondary forest showed significantly greater RLD, RSD and RVD than plantation forests. Between the two plantation types (PP and MF), the root density parameters showed no significant differences in RLD, RSD, or RVD across most soil depths. The only exception occurred at 20 cm depth, where PP shows significantly higher RSD ($0.24 \text{ cm}^2 \cdot \text{cm}^{-3}$) compared with MF ($0.18 \text{ cm}^2 \cdot \text{cm}^{-3}$).

3.2. Preferential path characteristics of typical vegetation types

The number of preferential paths and the dimensions of their infiltration aperture radii (R) varied in different vegetation types and soil depths (Fig. 6). The preferential paths in the typical vegetation cover of the Loess Plateau were primarily concentrated within the 0–15 cm soil layer, and had the highest number of preferential paths with R below 5 mm. The number of preferential paths in PP, MF and SF decreased with the increase in soil depth. Meanwhile, the number of preferential paths in natural secondary forests was overall greater than in plantation forests, with SF containing a total of 2524 preferential paths in 0–25 cm of soil, 1.39 times greater than PP and 1.48 times greater than MF. SF also demonstrated higher proportions of larger preferential paths ($R > 5 \text{ mm}$). At 5, 10 and 15 cm soil depths, SF's proportions of $R > 5 \text{ mm}$ paths were 34.8 %, 36.3 % and 35.2 % respectively, compared to plantation values of 23.5 %, 28.3 %, 24.3 % (PP) and 25.3 %, 29.5 %, 23.9 % (MF). Below a 15 cm soil depth, SF had fewer total preferential paths than plantations. However, the proportion of paths with $R > 5 \text{ mm}$ in SF was still relatively higher, at 27.3 % and 33.3 % at depths of 20 and 25 cm in the soil compared with 23.1 % and 10.6 % in PP, and 32.13 % and 7.55 % in MF, respectively.

In both plantations, the number of preferential paths was higher in PP than in MF at the 5 cm soil layer and lower in PP than in MF at the 10 and 15 cm soil layers. Meanwhile, the number of preferential paths showed a minimal difference in the two plantations in the deeper soil layers. The proportion of preferential paths with R of 5 mm or more is similar in PP and MF. Moreover, woodland had more preferential paths

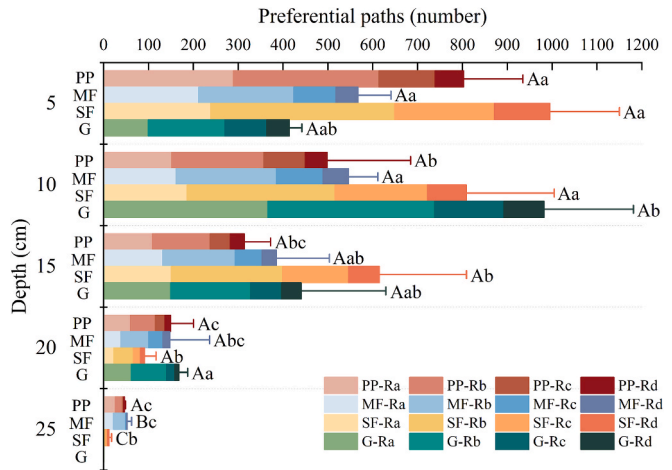


Fig. 6. Number of preferential paths in the soil for the different infiltration aperture radii (Ra, Rb, Rc and Rd refer to the preferential paths with infiltration aperture radii $1 \leq R < 2.5$, $2.5 \leq R < 5$, $5 \leq R < 10$ and $10 \leq R$, respectively) and vegetation types (PP, MF, SF and G). The error line is the standard deviation between the totals of the preferential paths. The significant differences are the results of the variations in the total number of preferential paths under different conditions; the different capital letters represent the significant differences in the total number of preferential paths for the various vegetation types at the same soil depth; the different lowercase letters represent the significant differences in the total number of preferential paths at various soil depths for the same vegetation type.

in 5 cm soil compared with grassland. At the 10 cm soil layer, grassland had a higher number of preferential paths than the three woodlands. Meanwhile, at soil layers below 10 cm, grassland still had a higher number of preferential paths than PP and MF, but lower than SF.

The planar coordinates of the preferential paths on the soil profile were obtained through image processing, and the distribution map was plotted, which can intuitively reveal the distribution characteristics of the soil preferential paths under the typical vegetation restoration mode (Fig. 7). Considering the large number of images, only one sample plot

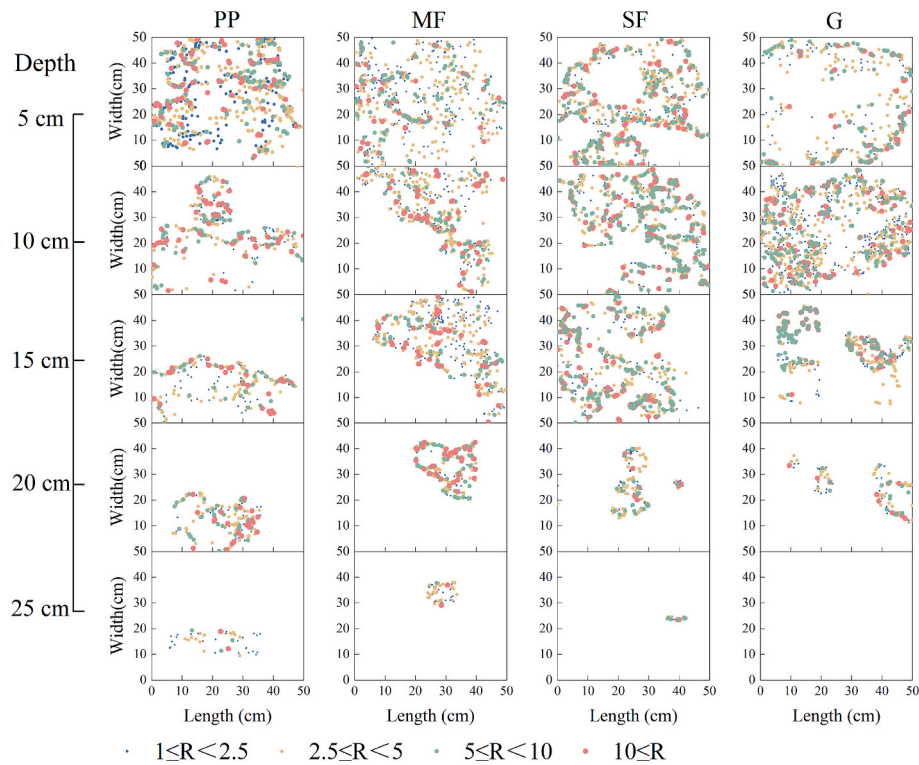


Fig. 7. Preferential path distribution in the soil.

under each vegetation cover was selected for display in this study.

The spatial distribution of preferential paths exhibited heterogeneity across soil layers for any vegetation type. The distribution of the preferential paths with the change of depth was not simply changed from large aggregates to smaller ones or from concentrated distribution to dispersed distribution, the entire distribution pattern and morphological configuration of the preferential paths also changed.

3.3. Contribution of roots on preferential paths

A certain polynomial relationship between the density of the roots and the number of preferential paths can be determined by summarising the RLD, RSD and RVD of roots and the preferential paths separately for all soil layers (Fig. 8). Overall, the roots in soil of the typical vegetation of the Loess Plateau play a contributing role in the formation of preferential paths.

Roots of different diameter classes acted together to form preferential paths of varying aperture radii in the soil. Consequently, the magnitude of the root contribution to preferential paths varied according to the diameter class of the roots (Fig. 9). Finer roots were more strongly correlated with the number of preferential paths. Roots above 4 mm in diameter had no significant effect ($p < 0.05$) on the preferential

paths of different aperture radii.

In the case of preferential paths of a given aperture size, the larger the diameter class of the roots, the smaller the influence of the RLD, RSD and RBD of the roots. The RLD, RSD and RBD of a root system for a given root diameter had less influence on preferential paths with R of less than 2.5 mm compared with those with R of more than 2.5 mm.

4. Discussion

4.1. Effects of vegetation on roots and preferential paths in the Loess Plateau

Vegetation restoration on the Loess Plateau has not only increased forest cover, but also promoted the growth of understory herbs and shrubs, which together form a complex root network structure in the soil (Guan et al., 2024a; Niu et al., 2024). Fine roots with smaller diameters, such as fibrous roots, concentrate in shallow soil layers; by contrast, thicker roots, such as the taproots of shrubs and certain herbaceous plants, can penetrate deeper into the soil. (Kulmatiski et al., 2020). In addition, the root distribution varied depending on the vegetation restoration mode. In natural secondary forests, the complex composition of plant species results in the presence of more roots in the soil than in

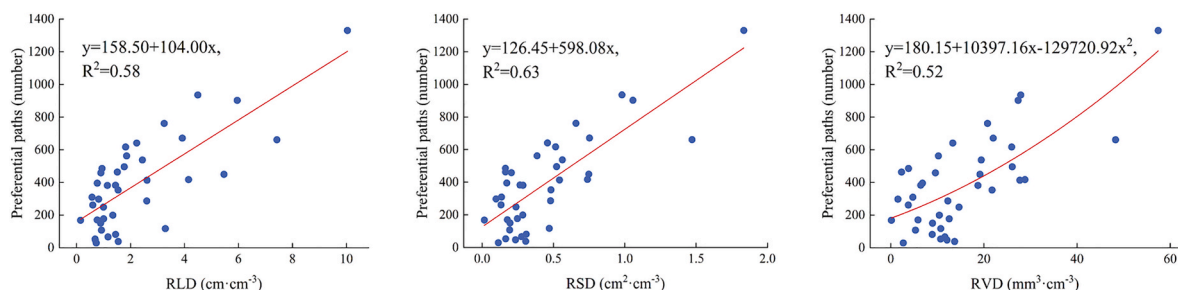


Fig. 8. Relationship between roots and preferential paths.

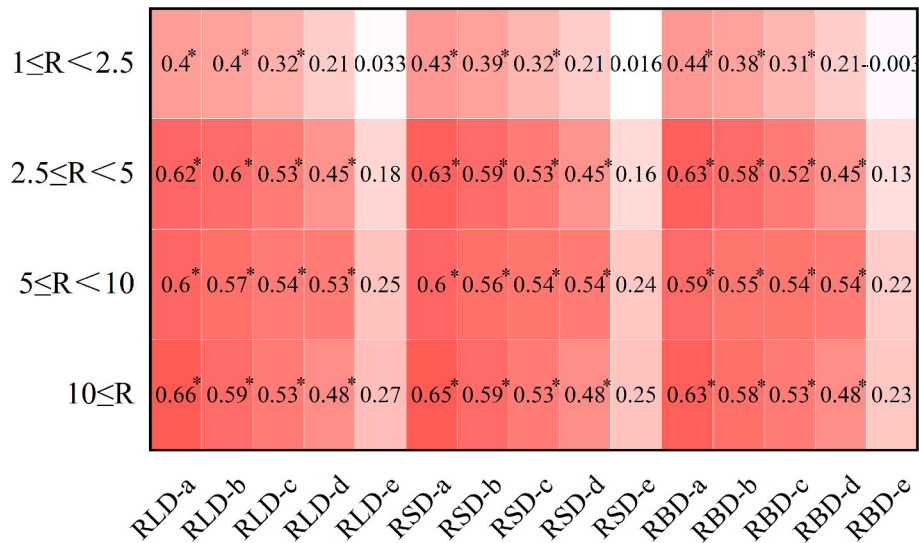


Fig. 9. Correlation between roots for different diameters (a, b, c, d and e refer to roots for $0 < D \leq 1$, $1 < D \leq 2$, $2 < D \leq 3$, $3 < D \leq 4$ and $4 < D$, respectively) and preferential paths for the various infiltration aperture radii. ** indicates a significant correlation ($p < 0.05$), where red denotes a positive correlation. The darker colour indicates a stronger correlation.

planted forests, especially in shallow soil.

A number of field investigations have shown that the root network structure present in the soil forms numerous preferential paths (Simone et al., 2020; Mair et al., 2023). These preferential flow paths form a variety of connectivity networks in the soil, which have varying degrees of influence on water and solute infiltration, transport and adsorption (Li et al., 2019a, 2019b). Although some studies have suggested that certain roots inhibit the development of preferential flow (Beven and Germann, 1982) this study aligns with the majority of research, indicating that both coarse and fine roots were able to promote the formation of preferential paths (Schwärzel et al., 2012). In particular, fine roots of smaller diameter classes often have well-developed shallow root systems and high lateral sprouting capacity, resulting in high heterogeneity of soil pore channels and providing good external conditions for the formation of preferential paths (Hou et al., 2021).

The number and distribution of preferential paths varied according to the vegetation restoration mode, which was closely related to the differences in root distribution in the soil. The secondary forest had a greater number of preferential paths in the soil than plantation forests due to their relatively high root density, and these paths also had a larger average infiltration aperture radius. In comparison with forest soil in other typical regions, such as karst areas of southwest China, the Loess Plateau forest soil exhibited fewer preferential paths but higher root density (Guan et al., 2023a; Guan et al., 2023b). This difference is likely due to the lower gravel content and fewer cracks in the Loess Plateau soil compared with the karst region (Finke et al., 2013). When compared with other land use types, such as highway slope soil without vegetative roots, the Loess plateau forest soil demonstrated more preferential paths (Zhang et al., 2017). This notion further suggests that root systems may enhance the formation of preferential paths.

However, the number of preferential paths was higher for all three typical vegetation covers than for grassland only in soil above 5 cm. This phenomenon suggests that forest restoration is more on enhancing the connectivity of preferential paths to the surface rather than merely increasing their number. Previous investigations in the present study area found that the degree of preferential flow development was lower in grassland than in woodland, whereas the degree of preferential flow development was higher in secondary forests than in plantation forests (Guan et al., 2024a). This notion suggests that the development degree of preferential flow depends not only on the number of preferential paths but also on their connectivity to the ground surface. If the

preferential paths have low connectivity to the surface, then a smaller water potential gradient can impede preferential flow infiltration, limiting the effectiveness of preferential flow paths in deeper soil layers (Guan et al., 2024b).

4.2. Implications of hydrological research and vegetation management on the Loess Plateau

Given the wind-driven depositional genesis of loess, affected by climatic conditions, it is subjected to varying degrees of weathering and loam-forming modification. Consequently, loess exhibits sub-stable structural characteristics, including non-homogeneous anisotropy, a loose and porous texture, columnar joint development and weak cementation (Li et al., 2024). This structure directly affects the hydrological phenomena in the Loess Plateau. In the later stage, the soil in the Loess Plateau develops preferential paths consisting of large pores, fissures and other structures under the influence of vegetation root growth. With the vegetation restoration programme on the Loess Plateau, more preferential paths have emerged in the soil, and the pattern of precipitation infiltration has changed considerably as a result.

The Philip model, Kostikov model, G-A model are commonly used to simulate the infiltration process in the Loess Plateau (Cheng et al., 2018). However, most of the infiltration models are unable to differentiate the movement by preferential flow from matrix infiltration. In the current context of vegetation restoration, preferential flow is an inevitable aspect of infiltration, and it significantly affects soil water recharge and surface flow on the Loess Plateau. Preferential flow is effective in recharging deep soil water (Clothier et al., 2010) and has a mitigating effect on the desiccation of the deep soil existing in the Loess Plateau. However, in some slopes with loose soil, precipitation may rapidly infiltrate along preferential paths to potential sliding surfaces deep in the soil, potentially leading to slope destabilisation (Tu et al., 2009). Accordingly, preferential flow is an essential aspect that cannot be ignored when analysing and modelling hydrological processes in the Loess Plateau. In addition to the role of the vegetation root system, external factors, such as soil faunal activity, freeze–thaw cycles, wet–dry alternation and hydrochemical solubility, can lead to the formation of preferential paths. The preferential flow network constituted by the preferential paths in the soil of the Loess Plateau is a complex hydraulic system (Chen et al., 2011). Therefore, extensive follow-up research is needed to support and clarify the interrelationships amongst the various

items of the system, and attention should be paid to water exchange, especially after rainfall, to reduce water (and even nutrient) leakage.

In this study, the number of preferential flow paths in the soil of monoculture *P. tabuliformis* and mixed *R. pseudoacacia*–*P. tabuliformis* forests was generally lower than that in natural secondary forest. This notion suggests that artificial plantations on the Loess Plateau still significantly lag behind natural secondary forests in facilitating water infiltration and soil moisture replenishment. Accordingly, this study proposes the adoption of close-to-nature afforestation approaches in future forestry development initiatives for the region. This strategy aims to restore stand structures and understory community characteristics resembling those of natural (secondary) forests, thereby ensuring rational and efficient utilisation of natural resources whilst maximising ecological restoration benefits.

4.3. Limitations of the study

Previous research in the study area have shown that the root system in soil was the main factor influencing the degree of preferential flow in the Loess Plateau woodland (Guan et al., 2024a). In order to extensively understand the role of the roots on preferential flow, this study conducted investigations and analyses focusing on preferential paths, which have received less attention from current research. Moreover, this study delineated the configuration of roots to explore their effects on the formation of preferential paths. The experimental design of this study has some limitations that need to be addressed and can provide guidance for future research.

(i) This study utilized the Brilliant Blue staining tracer method to trace preferential flow (and preferential paths), a straightforward in-situ technique that provides relatively accurate, direct visualisation of preferential flow across soil profiles, facilitating subsequent preferential path extraction. However, although Brilliant Blue is a widely used tracer, it also exhibits some deficiencies, such as problematic sorption and lower mobility than pure water in small pores (Nobles et al., 2010). Furthermore, this study conducted staining image analysis and processing methods, enabling straightforward and intuitive determination of preferential path quantities, infiltration aperture radius, and their spatial distribution in soil profiles. However, this method evaluated all blue stained areas as one colour. In reality, the soil profile may not be uniformly stained, with light blue indicating that only a small amount of water flowed by preferential path, and dark colours can show that a large amount of water flowed in preferential path. Future research could consider using other methods to obtain preferential paths (e.g. CT scanning and a ground-penetrating radar) to accurately characterise the preferential paths in soil.

(ii) In this study, the correlation between root density and the number of preferential paths was analysed to determine the specific role of roots on preferential paths. It was found that fine roots contributed more to the number of preferential paths and were more able to promote the formation of preferential paths. However, in addition to the number of preferential paths, morphological and structural parameters, such as curvature and connectivity can have significantly influence on infiltration and preferential flow (Lissy et al., 2020). Stronger root systems, compared with plants with only fine roots, can create more important channels around roots (biopores) in some cases, greatly facilitating preferential flow into infiltration (Vinatier et al., 2024). This phenomenon may be precisely related to the structure of the preferential paths. Future studies could focus on the 3D morphology of preferential paths and further explore the mechanisms of preferential flow, including the influence of the root system or other environmental factors on this process.

5. Conclusion

By investigating and analysing roots and preferential flow paths in soil of typical vegetation types of the Loess Plateau, this study aims to

investigate the influence of roots on the formation of preferential paths in the process of ‘vegetation–roots–infiltration’, and to provide a basis for subsequent research on preferential paths and preferential flow.

The results indicated that vegetation cover increased the density (root length density, root surface density and root volume density) of roots in the soil and the number of preferential paths in the top 0–5 cm of soil, particularly in the natural secondary forest. Preferential paths for infiltration formed by the roots of forest and grass vegetation were predominantly distributed in the 0–15 cm soil layer. Most, preferential paths had a staining infiltration aperture radius less than 5 mm. Roots of varying diameters contributed to the formation of preferential paths, but the effect of fine roots was more pronounced than that of coarse roots. This study broadens the scope of research on soil preferential flow phenomena in the Loess Plateau and provides new insights into how vegetation restoration affects the formation of soil preferential paths. Furthermore, this study contributes to the understanding of the phenomenon of ‘vegetation restoration – change in root density – change in soil preferential paths – change in precipitation infiltration characteristics’. The following are some of the issues that need to be further investigated: (i) the effect of vegetation restoration on the 3D structure of preferential paths (e.g. curvature and connectivity) and (ii) hydrological modelling that explains the flow of infiltrated water along the preferential paths (i.e. preferential flow transport mechanisms).

CRediT authorship contribution statement

Ning Guan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Huaxing Bi:** Supervision, Conceptualization. **Yilin Song:** Investigation. **Shanhong Lu:** Investigation. **Dandan Lin:** Investigation. **Jindan Han:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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